NEUROSEG™ CLINICAL NEURO-ONCOLOGY AI REPORT

Automated Multi-Parametric MRI Tumor Sub-Region Volumetrics & RANO Criteria Quantification

Patient / Scan ID:	**BraTS2021_00001**	**Report Date:**	2026-10-17 05:05
Sequences:	T1, T1CE, T2, FLAIR	**Spatial Resampling:**	1.0 mm³ Isotropic (RAS)
Mask Source:	Not specified	**Post-Processing:**	3D Connected-Component Cleaned

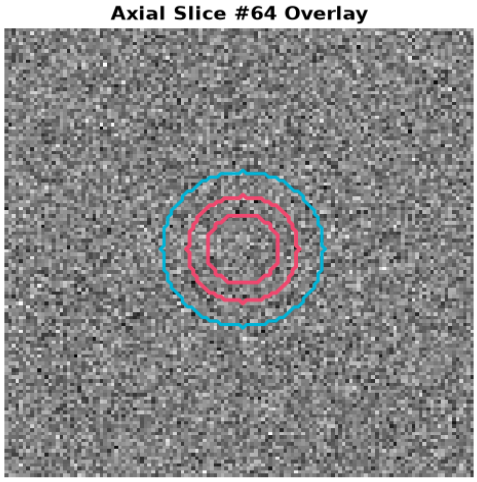
1. Quantitative Tumor Volumetrics & Compartment Distribution

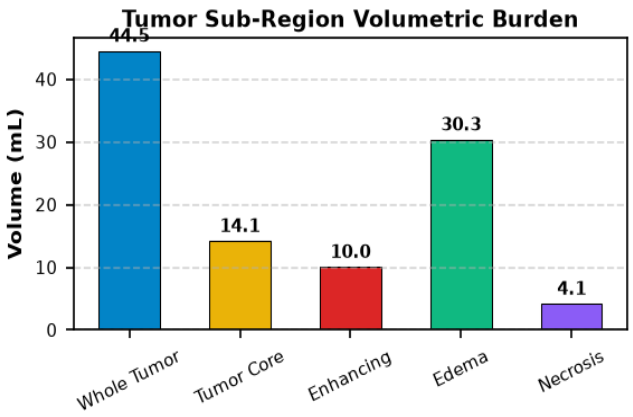
Tumor Compartment	Raw Voxels	Cleaned Voxels	Filtered Noise	Volume (mm³)	Volume (mL)
Whole Tumor (WT)	44,473	44,473	0	44,473.0	**44.473 mL**
Tumor Core (TC)	14,147	14,147	0	14,147.0	**14.147 mL**
Enhancing Tumor (ET)	10,008	10,008	0	10,008.0	**10.008 mL**
Peritumoral Edema	-	Derived	-	30,326.0	**30.326 mL**
Necrotic Core / Cavity	-	Derived	-	4,139.0	**4.139 mL**

2. RANO (Response Assessment in Neuro-Oncology) Criteria Evaluation

Target Enhancing Diameter (D1):	**28.84 mm**	**Perpendicular Diameter (D2):**	**28.84 mm**
Sum of Products (SPD):	**832.0 mm²**	**Measurement Slice Index:**	Axial Slice #60
Lesion Measurability:	**Measurable Lesion (>= 10x10 mm)**	**Standard Criteria:**	RANO 2D Bidirectional Threshold (>= 10x10 mm)

3. Multi-Slice Visual Overlay & Compartment Analytics





4. Clinical Impression & Neurosurgical Guidance

Multi-parametric 3D MRI segmentation reveals a total Whole Tumor volume of **44.47 mL**, with an active contrast-enhancing tumor margin of **10.01 mL** and associated peritumoral edema measuring **30.33 mL**. Connected-component cleaning removed **0** false-positive voxels. RANO 2D cross-product is **832.0 mm²** (Measurable Lesion (>= 10x10 mm)). This automated output is for research use only. It requires review by a qualified neuroradiologist and is not a clinical recommendation.

Validation status: Research prototype - not clinically validated

Software: NeuroSeg research prototype (MONAI 1.6+)

Attending Neuroradiologist: _____

Signature / Approval: _____